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Semiconductor device with semiconductor element and plural first and second leads

Abstract

A semiconductor device includes an electric conductor, a semiconductor element, and a bonding layer. The electric conductor has a main surface and a rear surface opposite to the main surface in a thickness direction. The semiconductor element includes a main body and electrodes. The main body has a side facing the main surface of the conductor, and the electrodes each protrude toward the main surface from the side of the main body to be electrically connected to the main surface. The bonding layer is held in contact with the main surface and the electrodes. Each electrode includes a base portion in contact with the main body, and a columnar portion protruding toward the main surface from the base portion to be held in contact with the bonding layer, which is a sintered body of a metal powder.

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Background/Summary

FIELD

(1) The present disclosure relates to a semiconductor device in which a semiconductor element is electrically bonded to conductive members through flip-chip bonding.

BACKGROUND

(2) Semiconductor devices in which a semiconductor element is electrically bonded to conductive members (leads or the like) through flip-chip bonding have conventionally been widely known. JP-A-2018-85522 discloses an example of such a semiconductor device.

(3) In the conventional semiconductor device, multiple electrodes of the semiconductor element (in JP-A-2018-85522, a semiconductor chip) are electrically bonded to conductive members (drawn-out wires) by a bonding layer (conductive bumps). The multiple electrodes of the semiconductor element oppose the conductive members.

(4) In general, solder is often used for the bonding layer. For example, as the melting point of lead-free solder is 217° C., solder is a conductor with a relatively low melting point. In this case, when the semiconductor apparatus is used for a long period, electromigration may occur in the bonding layer using the solder. Electromigration tends to occur in conductors with a relatively low melting point. Electromigration is a phenomenon in which loss such as a void is generated in a conductor due to ions gradually moving since momentum is exchanged between the electrons moving in the conductor and the metal atoms.

(5) Accompanying a reduction in the size of semiconductor elements and an increase in integration of circuits formed inside of a semiconductor element in recent years, multiple electrodes of a semiconductor element have tended to decrease in size and increase in number. When electromigration occurs in a bonding layer used to bond these multiple electrodes and conductive members, a decrease in the strength with which the multiple electrodes are bonded to the conductive members and an increase in the parasitic resistance of the semiconductor apparatus cannot be ignored. Note that electromigration is more likely to occur also due to the current concentration in the conductor increasing.

SUMMARY

(6) In view of the above-described circumstances, the present disclosure aims to provide a semiconductor device that can suppress the occurrence of electromigration in a bonding layer used for flip-chip bonding.

(7) According to an aspect of the present disclosure, there is provided a semiconductor device that includes: an electric conductor having a main surface and a rear surface that face mutually opposite sides in a thickness direction; a semiconductor element including an element main body and a

plurality of electrodes, where the element main body has a first side facing the main surface of the conductor, and the plurality of electrodes each protrude toward the main surface from the first side of the element main body to be electrically connected to the main surface; and a bonding layer held in contact with the main surface and the plurality of electrodes. Each of the plurality of electrodes includes: a base portion in contact with the element main body; and a columnar portion protruding toward the main surface from the base portion and held in contact with the bonding layer. The bonding layer is a sintered body of a metal powder.

(8) Preferably, the columnar portion of each electrode includes a leading end surface facing the main surface and a side surface connected to the leading end surface and facing in a direction orthogonal to the thickness direction, where the bonding layer is in contact with the leading end surface and the side surface.

(9) Preferably, the semiconductor element includes an outer surface protection film that covers the first side of the element main body, and the leading end surface of each electrode is located between the main surface and the outer surface protection film in the thickness direction.

(10) Preferably, the outer surface protection film is in contact with the base portion and the columnar portion of each electrode.

(11) Preferably, the outer surface protection film is spaced apart from the columnar portion of each electrode.

(12) Preferably, the columnar portion of each electrode is provided with a recessed portion that is recessed toward the element main body from the leading end surface, and the bonding layer is in contact with the recessed portion.

(13) Preferably, the leading end surface of each electrode has a protruding shape that bulges toward the main surface.

(14) Preferably, the base portion of each electrode is made of a material including aluminum, or a material including copper.

(15) Preferably, the metal powder includes silver or copper.

(16) Preferably, the electric conductor includes: a plurality of first leads that extend in a first direction orthogonal to the thickness direction and are separated from each other in a second direction orthogonal to the thickness direction and the first direction; and a plurality of second leads that are located offset from the plurality of first leads in the second direction. The semiconductor element includes a semiconductor substrate and a semiconductor layer disposed on the semiconductor substrate and facing the main surface. The semiconductor layer is formed with a switching circuit and a control circuit electrically connected to the switching circuit. The plurality of electrodes include a plurality of first electrodes and a plurality of second electrodes. The plurality of first electrodes are electrically connected to the switching circuit and are electrically connected to the plurality of first leads. The plurality of second electrodes are electrically connected to the control circuit, and at least part of the plurality of second electrodes is electrically connected to the plurality of second leads.

(17) Preferably, the semiconductor device further includes a sealing resin that covers a part of each first lead, a part of each second lead and the semiconductor element. The sealing resin has: a bottom surface that faces a same side as the rear surface in the thickness direction; and a pair of first side surfaces that are connected to the bottom surface and are spaced apart from each other in the first direction. The plurality of first leads each include a main portion that extends in the first direction and a pair of side portions that are connected to ends of the main portion in the first direction. Each of the pair of side portions has a first end surface that faces in the first direction. The rear surface of each first lead is exposed from the bottom surface. The first end surfaces are exposed from the pair of first side surfaces so as to be flush with the first side surfaces. The dimension in the second direction of the pairs of first end surfaces is smaller than the dimension in the second direction of the rear surfaces of the main portions.

(18) Preferably, each of the pair of side portions is provided with wedge portions that extend from

the main surface to the rear surface and are recessed inward of the side portions from both sides in the second direction.

(19) Preferably, each of the pair of side portions is provided with a notch portion that extends from the main surface to the rear surface, is recessed in the first direction from the first end surface, and divides the first end surface into two regions.

(20) Preferably, each of the plurality of second leads includes a second end surface facing in the second direction. The sealing resin includes a pair of second side surfaces that are connected to the bottom surface and the pair of first side surfaces and are spaced apart from each other in the second direction. The rear surfaces of the plurality of second leads are exposed from the bottom surface. The plurality of second end surfaces are exposed from one of the second side surfaces so as to be flush with the one of the second side surfaces.

(21) Preferably, one of the plurality of first leads is formed with a plurality of protruding portions that protrude from the main portion of the one of the plurality of first leads. Each of the plurality of protruding portions includes an auxiliary end surface that faces in the second direction. The auxiliary end surface of each protruding portion is exposed from the other of the second side surfaces so as to be flush with the other of the second side surfaces.

(22) According to the semiconductor device of the present disclosure, it is possible to suppress the occurrence of electromigration in a bonding layer used in flip-chip bonding.

(23) Other features and advantages of the present disclosure will become more evident through detailed description given below with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a semiconductor device according to a first embodiment.
- (2) FIG. 2 is a plan view of the semiconductor device shown in FIG. 1 (sealing resin is shown transparent).
- (3) FIG. 3 is a plan view of the semiconductor device shown in FIG. 1 (a semiconductor element and sealing resin are shown transparent).
- (4) FIG. 4 is a bottom view of the semiconductor device shown in FIG. 1.
- (5) FIG. 5 is a front view of the semiconductor device shown in FIG. 1.
- (6) FIG. 6 is a rear view of the semiconductor device shown in FIG. 1.
- (7) FIG. 7 is a right-side view of the semiconductor device shown in FIG. 1.
- (8) FIG. 8 is a left-side view of the semiconductor device shown in FIG. 1.
- (9) FIG. 9 is a partially enlarged view of FIG. 3.
- (10) FIG. 10 is a partially enlarged view of FIG. 3.
- (11) FIG. 11 is a cross-sectional view taken along line XI-XI in FIG. 3.
- (12) FIG. 12 is a cross-sectional view taken along line XII-XII in FIG. 3.
- (13) FIG. 13 is a cross-sectional view taken along line XIII-XIII in FIG. 3.
- (14) FIG. 14 is a cross-sectional view taken along line XIV-XIV in FIG. 3.
- (15) FIG. 15 is a partially enlarged view of FIG. 11 (near a first electrode).
- (16) FIG. 16 is a partially enlarged view of FIG. 11 (near a second electrode).
- (17) FIG. 17 is a partially enlarged cross-sectional view of a semiconductor device according to a second embodiment (near a first electrode).
- (18) FIG. 18 is a partially enlarged cross-sectional view of the semiconductor device shown in FIG. 17 (near a second electrode).
- (19) FIG. 19 is a partially enlarged cross-sectional view of a semiconductor device according to a third embodiment (near a first electrode).
- (20) FIG. 20 is a partially enlarged cross-sectional view of the semiconductor device shown in FIG.

19 (near a second electrode).

EMBODIMENTS

(21) Embodiments of the present disclosure will be described with reference to the accompanying drawings.

(22) A semiconductor device **A10** according to a first embodiment will be described with reference to FIGS. **1** to **16**. The semiconductor device **A10** includes conductive members **10**, a semiconductor element **20**, a bonding layer **30**, and sealing resin **40**. As shown in FIG. **1**, the package format of the semiconductor device **A10** is QFN (Quad For Non-Lead Package). The semiconductor element **20** is a flip-chip LSI. A switching circuit **212A** and a control circuit **212B** (each of which will be described in detail later) are included inside of the semiconductor element **20**. In the semiconductor device **A10**, direct-current electrical power (voltage) is converted into alternating-current electrical power (voltage) by the switching circuit **212A**. The semiconductor device **A10** is used as an element included in a circuit of a DC/DC converter, for example. Here, in FIG. **2**, the sealing resin **40** is shown transparent to facilitate understanding. In FIG. **3**, the semiconductor element **20** and the sealing resin **40** are shown transparent in order to facilitate understanding. In these drawings, the semiconductor element **20** and sealing resin **40**, which are shown transparent, are indicated by virtual lines (two-dot chain lines).

(23) In the description of the semiconductor device **A10**, a thickness direction **z** of the conductive member **10** may be referred to as “thickness direction **z**”. The direction orthogonal to the thickness direction **z** may be referred to as “first direction **x**”. The direction orthogonal to both the thickness direction **z** and the first direction **x** may be referred to as “second direction **y**”. As shown in FIGS. **1** and **2**, the semiconductor device **A10** has a square shape in a view in the thickness direction **z**. Also, in the description of the semiconductor device **A10**, for the sake of convenience, the side in the second direction **y** on which multiple second leads **12** (described in detail later) are located will be referred to as “one side in the second direction **y**”. The side in the second direction **y** on which multiple first leads **11** (described in detail later) are located will be referred to as “other side in the second direction **y**”.

(24) As shown in FIG. **2**, the conductive members **10** are terminals for supporting the semiconductor element **20** and mounting the semiconductor device **A10** on a wiring substrate. As shown in FIGS. **11** to **14**, some of the conductive members **10** are covered by the sealing resin **40**. The conductive members **10** each have a main surface **101** and a rear surface **102** that face mutually opposite directions in the thickness direction **z**. The main surfaces **101** face one side in the thickness direction **z** and oppose the semiconductor element **20**. The semiconductor element **20** is supported by the main surfaces **101**. The main surfaces **101** are covered by the sealing resin **40**. The rear surfaces **102** face the other side in the thickness direction **z**. The conductive members **10** are formed from a single lead frame. The constituent material of the lead frame is, for example, copper (Cu) or a copper alloy. The conductive members **10** include multiple first leads **11**, multiple second leads **12**, and a pair of third leads **13**.

(25) As shown in FIGS. **3** and **4**, the multiple first leads **11** have band shapes that extend in the second direction **y** in a view in the thickness direction **z**. The multiple first leads **11** are aligned in the second direction **y**. In the example indicated by the semiconductor apparatus **A10**, the multiple first leads **11** are constituted by three terminals, namely a first input terminal **11A**, a second input terminal **11B**, and an output terminal **11C**. The multiple first leads **11** are arranged in the following order from the one side to the other side in the second direction **y**: the first input terminal **11A**, the output terminal **11C**, the second input terminal **11B**. The first input terminal **11A** and the second input terminal **11B** receive input of direct-current electrical power (voltage), which is to be subjected to power conversion in the semiconductor apparatus **A10**. The first input terminal **11A** is a cathode (P terminal). The second input terminal **11B** is an anode (N terminal). The output terminal **11C** outputs an alternating-current electrical power (voltage) resulting from power conversion performed by the switching circuit **212A** included in the semiconductor element **20**.

(26) As shown in FIG. 3, the first input terminal **11A** is located between the multiple second leads **12** and the output terminal **11C** in the second direction *y*. The output terminal **11C** is located between the first input terminal **11A** and the second input terminal **11B** in the second direction *y*. The first input terminal **11A** and the output terminal **11C** each include a main portion **111** and a pair of side portions **112**. As shown in FIGS. 3 and 4, the main portion **111** extends in the first direction *x*. In the multiple first leads **11**, the semiconductor element **20** is supported by the main surfaces **101** of the main portions **111**. The pair of side portions **112** are connected to both ends in the first direction *x* of the main portion **111**. As shown in FIGS. 3, 4, 12, and 13, each of the pairs of side portions **112** has first end surfaces **112A**. The first end surfaces **112A** are connected to both the main surfaces **101** and the rear surfaces **102** of the first leads **11**, and face in the first direction *x*. The first end surfaces **112A** are exposed from the sealing resin **40**.

(27) As shown in FIG. 9, wedge portions **112B** are formed on each of the pairs of side portions **112** of the first input terminal **11A** and the output terminal **11C**. The wedge portions **112B** extend from the main surfaces **101** to the rear surfaces **102** of the first leads **11** and are recessed inward of the side portions **112** from both sides in the second direction *y*. The wedge portions **112B** are in contact with the sealing resin **40**. Due to the wedge portions **112B**, in the first input terminal **11A** and the output terminal **11C**, a dimension *b* in the second direction *y* of each of the pair of first end surfaces **112A** is smaller than a dimension *B* in the second direction *y* of the rear surface **102** of the main portion **111**.

(28) As shown in FIG. 3, the second input terminal **11B** is located on the other side in the second direction *y* with respect to the output terminal **11C**. For this reason, the second input terminal **11B** is located on the other side in the second direction *y* among the multiple first leads **11**. The second input terminal **11B** includes a main portion **111**, a pair of side portions **112**, and multiple protruding portions **113**. The multiple protruding portions **113** protrude from the other side in the second direction *y* of the main portion **111**. The sealing resin **40** fills the spaces between every two adjacent protruding portions **113**. As shown in FIG. 12, the multiple protruding portions **113** each have an auxiliary end surface **113A**. The auxiliary end surfaces **113A** are connected to both the main surface **101** and the rear surface **102** of the second input terminal **11B**, and face the other side in the second direction *y*. The auxiliary end surface **113A** is exposed from the sealing resin **40**. As shown in FIG. 7, the multiple auxiliary end surfaces **113A** are aligned at a predetermined interval along the first direction *x*.

(29) As shown in FIG. 10, notch portions **112C** are formed on the pair of side portions **112** of the second input terminal **11B**. The notch portions **112C** extend from the main surface **101** to the rear surface **102** of the second input terminal **11B** and are recessed in the first direction *x* from the first end surface **112A**. Accordingly, the first end surface **112A** is divided into two regions spaced apart from each other in the second direction *y*. Due to the notch portions **112C** as well, the dimension *b* in the second direction *y* of each of the pair of first end surfaces **112A** is smaller than the dimension *B* in the second direction *y* of the rear surface **102** of the main portion **111** in the second input terminal **11B**. Note that the dimension *b* in this context is obtained by adding together a dimension *b1* in the second direction *y* of one region of the first end surface **112A** and a dimension *b2* in the second direction *y* of another region of the first end surface **112A** ($b=b1+b2$). The sealing resin **40** fills the notch portions **112C**.

(30) As shown in FIGS. 3 and 4, the area of the main surface **101** is greater than the area of rear surface **102** in each of the multiple first leads **11**. In the example indicated by the semiconductor device **A10**, the areas of the rear surfaces **102** of the first input terminal **11A** and the output terminal **11C** are equal to each other. The area of the rear surface **102** of the second input terminal **11B** is greater than the area of each rear surface **102** of the first input terminal **11A** and the output terminal **11C**.

(31) In each of the first input terminal **11A**, the second input terminal **11B**, and the output terminal **11C**, for example, silver (Ag) plating may be performed on the main surface **101** of the main

portion **111** supported by the semiconductor element **20**. Furthermore, in each of the first input terminal **11A**, the second input terminal **11B**, and the output terminal **11C**, for example, tin (Sn) plating may be performed on the rear surface **102**, the pair of first end surfaces **112A**, and the multiple auxiliary end surfaces **113A** that are exposed from the sealing resin **40**. Note that instead of tin plating, multiple types of metal plating, in which nickel (Ni), palladium (Pd), and gold (Au) are stacked in the stated order, may also be used.

(32) As shown in FIG. 3, the multiple second leads **12** are located on the one side in the second direction *y* with respect to the multiple first leads **11**. Any one of the multiple second leads **12** is a contact terminal of a control circuit **212B** included in the semiconductor element **20**. The other multiple second leads **12** receive electrical power (voltage) for driving the control circuit **212B**, or an electrical signal for transmitting to the control circuit **212B**. As shown in FIGS. 3, 4, and 11, the multiple second leads **12** each have a second end surface **121**. The second end surfaces **121** are connected to both the main surfaces **101** and the rear surfaces **102** of the second leads **12** and face the one side in the second direction *y*. The second end surfaces **121** are exposed from the sealing resin **40**. As shown in FIG. 8, the multiple second end surfaces **121** are aligned at a predetermined interval in the first direction *x*.

(33) As shown in FIGS. 3 and 4, the area of the main surface **101** is greater than the area of rear surface **102** in each of the multiple second leads **12**. Note that the areas of the rear surfaces **102** of the multiple second leads **12** are all equal to each other. For example, silver plating may be performed on the rear surfaces **102** of the multiple second leads **12** on which the semiconductor element **20** is supported. Furthermore, for example, tin plating may also be performed on the rear surfaces **102** and the second end surface **121** of the multiple second leads **12** exposed from the sealing resin **40**. Note that instead of the plating, for example, multiple types of metal plating in which nickel, palladium, and gold are stacked in the stated order may also be used.

(34) As shown in FIG. 3, the pair of third leads **13** are located between the first leads **11** (first input terminals **11A**) and the multiple second leads **12** in the second direction *y*. The pair of third leads **13** are spaced apart from each other in the first direction *x*. Each of the pair of third leads **13** receives input of an electrical signal for transmitting to the control circuit **212B** included in the semiconductor element **20**, and the like. As shown in FIGS. 3, 4, and 14, the pair of third leads **13** each have a third end surface **131**. The third end surfaces **131** are connected to both the main surfaces **101** and the rear surfaces **102** and face in the first direction *x*. The third end surfaces **131** are exposed from the sealing resin **40**. The third end surfaces **131** are aligned in the second direction *y* along with the first end surfaces **112A** of the multiple first leads **11**.

(35) As shown in FIGS. 3 and 4, the area of the main surface **101** is greater than the area of rear surface **102** in each of the pair of third leads **13**. For example, silver plating may also be performed on the main surfaces **101** of the pair of third leads **13** on which the semiconductor element **20** is supported. Furthermore, for example, tin plating may also be performed on the rear surfaces **102** and the third end surfaces **131** of the pair of third leads **13** exposed from the sealing resin **40**. Note that instead of tin plating, for example, multiple types of metal plating in which nickel, palladium, and gold are stacked in the stated order may also be used.

(36) As shown in FIGS. 11 to 14, the semiconductor element **20** is electrically bonded to and supported by the conductive member **10** (the multiple first leads **11**, the multiple second leads **12**, and the pair of third leads **13**) through flip-chip bonding. The semiconductor element **20** is covered by the sealing resin **40**. As shown in FIGS. 12 to 18, the semiconductor element **20** includes an element main body **21**, multiple electrodes **22**, and an outer surface protection film **23**.

(37) The element main body **21** forms the main portion of the semiconductor element **20**. As shown in FIGS. 15 and 16, the element main body **21** includes a semiconductor substrate **211**, a semiconductor layer **212**, and a passivation film **213**.

(38) As shown in FIGS. 15 and 16, the semiconductor substrate **211** supports the semiconductor layer **212**, the passivation film **213**, the multiple electrodes **22**, and the outer surface protection film

23 on its lower side. The constituent material of the semiconductor substrate **211** is, for example, Si (silicon) or silicon carbide (SiC).

(39) As shown in FIGS. **11** to **14**, the semiconductor layer **212** is stacked on the side of the semiconductor substrate **211** that opposes the main surfaces **101** of the conductive members **10**. The semiconductor layer **212** includes multiple types of p-type semiconductors and n-type semiconductors based on differences in the amount of doped elements. A switching circuit **212A** and a control circuit **212B** that is electrically connected to the switching circuit **212A** are included in the semiconductor layer **212**. The switching circuit **212A** is a MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor), an IGBT (Insulated Gate Bipolar Transistor), or the like. In the example indicated by the semiconductor device **A10**, the switching circuit **212A** is divided into two regions, namely a high-voltage region (upper arm circuit) and a low-voltage region (lower arm circuit). The regions are each constituted by one n-channel MOSFET. The control circuit **212B** includes a gate driver for driving the switching circuit **212A**, a bootstrap circuit corresponding to the high-voltage region of the switching circuit **212A**, and the like, and performs control for normally running the switching circuit **212A**. Note that a wiring layer (not shown) is included in the semiconductor layer **212**. The switching circuit **212A** and the control circuit **212B** are electrically connected to each other due to the wiring layer.

(40) As shown in FIGS. **15** and **16**, the passivation film **213** covers the lower surface of the semiconductor layer **212**. The passivation film **213** has an electrically insulating property. For example, the passivation film **213** is constituted by a silicon oxide film (SiO₂) that is in contact with the lower surface of the semiconductor layer **212**, and a silicon nitride film (Si₃N₄) that is stacked on the silicon oxide film. The passivation film **213** is provided with multiple openings **213A** that penetrate in the thickness direction **z**.

(41) As shown in FIGS. **11** to **14**, the multiple electrodes **22** protrude from the side of the element main body **21** opposing the main surfaces **101** of the conductive members **10**, toward the main surfaces **101** of the conductive members **10**. Note that the upper ends of the multiple electrodes **22** are in contact with the semiconductor layer **212** of the element main body **21**. The multiple electrodes **22** are electrically bonded to the main surfaces **101** of the conductive members **10**. The multiple electrodes **22** include multiple first electrodes **22A** and multiple second electrodes **22B**. The multiple first electrodes **22A** are electrically connected to the switching circuit **212A** of the semiconductor layer **212**. In addition, the multiple first electrodes **22A** are electrically bonded to the main surfaces **101** of the multiple first leads **11**. Accordingly, the multiple first leads **11** are electrically connected to the switching circuit **212A**. Also, the multiple second electrodes **22B** are electrically connected to the control circuit **212B** of the semiconductor layer **212**. In addition, a majority of the multiple second electrodes **22B** are electrically bonded to the main surfaces **101** of the multiple second leads **12**. In addition, the rest of the multiple second electrodes **22B** are electrically bonded to the main surfaces **101** of the pair of third leads **13**. Accordingly, the multiple second leads **12** and the pair of third leads **13** are electrically connected to the control circuit **212B**.

(42) As shown in FIGS. **15** and **16**, the multiple electrodes **22** each include a base portion **221** and a columnar portion **222**. Note that base portion **221** is in contact with the semiconductor layer **212** of the element main body **21**. Accordingly, the base portion **221** is electrically connected to one of the switching circuit **212A** and the control circuit **212B** of the semiconductor layer **212**. The constituent material of the base portion **221** includes aluminum (Al) or copper. In another configuration of the base portion **221**, multiple types of metal layers may also be used, in which copper, nickel, and palladium are stacked in the stated order downward from the semiconductor layer **212**. The base portion **221** is in contact with the passivation film **213** of the element main body **21**. A portion of the base portion **221** is exposed from the opening **213A** of the passivation film **213**. The columnar portion **222** protrudes toward the main surface **101** of the conductive member **10** from the portion of the base portion **221** exposed from the opening **213A**. The columnar portion **222** has a circular column shape, for example. The constituent material of the

columnar portion **222** includes copper. The columnar portion **222** includes a leading end surface **222A** and a side surface **222B**. The leading end surface **222A** opposes the main surface **101** of the conductive member **10**. The side surface **222B** is connected to the leading end surface **222A** and faces the direction orthogonal to the thickness direction *z*. In the semiconductor device **A10**, a recessed portion **222C** that is recessed from the leading end surface **222A** to the element main body **21** is formed in the columnar portion **222**. Note that the multiple electrodes **22** are formed through electroplating.

(43) As shown in FIGS. **15** and **16**, the outer surface protection film **23** covers the side of the element main body **21** that opposes the main surfaces **101** of the conductive members **10**, that is, the passivation film **213** of the element main body **21**. In each of the multiple electrodes **22**, the leading end surface **222A** of the columnar portion **222** is located between the main surface **101** of the conductive member **10** and the outer surface protection film **23** in the thickness direction *z*. In the semiconductor apparatus **A10**, the outer surface protection film **23** is in contact with both the base portion **221** and the columnar portion **222** of the multiple electrodes **22**. The outer surface protection film **23** has an electrically insulating property. The constituent material of the outer surface protection layer **23** is, for example, polyamide.

(44) As shown in FIGS. **15** and **16**, the bonding layer **30** is in contact with both the main surfaces **101** of the conductive members **10** and the multiple electrodes **22**. The bonding layer is conductive. Accordingly, the multiple electrodes **22** are electrically bonded to the main surfaces **101** of the conductive members **10**. The bonding layer **30** is a sintered body in which metal powder is bonded together. The metal powder includes silver or copper. The metal powder becomes a sintered body under a temperature condition of 200° C. to 300° C. According to this temperature condition, a lead-free solder will melt. For this reason, the bonding layer **30**, which is a sintered body, can be formed under a temperature condition similar to that in the case of melting a lead-free solder. In each of the multiple electrodes **22**, the bonding layer **30** is in contact with both the leading end surface **222A** and the side surface **222B** of the columnar portion **222**. In the semiconductor device **A10**, the bonding layer **30** is also in contact with the recessed portion **222C** of the columnar portion **222**.

(45) As shown in FIGS. **5** to **8**, the sealing resin **40** includes a peak surface **41**, a bottom surface **42**, a pair of first side surfaces **431**, and a pair of second side surfaces **432**. The constituent material of the sealing resin **40** is, for example, black epoxy resin.

(46) As shown in FIGS. **11** to **14**, the peak surface **41** faces the same side as the main surfaces **101** of the conductive members **10** in the thickness direction *z*. As shown in FIGS. **5** to **8**, the bottom surface **42** faces the side opposite to the peak surface **41**. As shown in FIG. **4**, the rear surfaces **102** of the multiple first leads **11**, the rear surfaces **102** of the second leads **12**, and the rear surfaces **102** of the pair of third leads **13** are exposed from the bottom surface **42**.

(47) As shown in FIGS. **7** and **8**, the pair of first side surfaces **431** are connected to both the peak surface **41** and the bottom surface **42** and face in the first direction *x*. The pair of first side surfaces **431** are spaced apart from each other in the second direction *y*. As shown in FIGS. **12** to **14**, the first end surfaces **112A** of the multiple first leads **11** and the third end surfaces **131** of the third leads **13** are exposed from the pair of first side surfaces **431** so as to be flush with each other.

(48) As shown in FIGS. **5** and **6**, the pair of second side surfaces **432** are connected to the peak surface **41**, the bottom surface **42**, and the pair of first side surfaces **431**, and face in the second direction *y*. The pair of second side surfaces **432** are spaced apart from each other in the first direction *x*. As shown in FIG. **11**, the second end surfaces **121** of the multiple second leads **12** are exposed from the second side surface **432** located on the one side in the second direction *y* so as to be flush with the second side surface **432**. The multiple auxiliary end surfaces **113A** of the second input terminals **11B** (first leads **11**) are exposed from the second side surface **432** located on the other side in the second direction *y* so as to be flush with the second side surface **432**.

(49) Next, actions and effects of the semiconductor device **A10** will be described.

(50) The semiconductor device A10 includes: conductive members 10 having main surfaces 101; a semiconductor element having an element main body 21 and multiple electrodes 22 that are electrically bonded to the main surfaces 101; and a bonding layer 30 that is in contact with both the main surfaces 101 and the multiple electrodes 22. The multiple electrodes 22 each have a base portion 221 that is in contact with the side of the element main body 21 opposing the main surface 101, and a columnar portion 222 that protrudes from the base portion 221 to the main surfaces 101 and is in contact with the bonding layer 30. Accordingly, the semiconductor element 20 is electrically bonded to the conductive member 10 through flip-chip bonding. The bonding layer 30 is a sintered body in which metal powder is bonded together. The melting point of the sintered body can be set to about 1000° C. (if the metal powder is silver or copper), which is close to five times the melting point of solder. Accordingly, with the semiconductor device A10, it is possible to suppress the occurrence of electromigration in the bonding layer 30 used in flip-chip bonding.

(51) The columnar portions 222 of the electrodes 22 of the semiconductor element 20 each include a leading end surface 222A that opposes the main surface 101 of the conductive member 10, and a side surface 222B. The bonding layer 30 is in contact with both the leading end surface 222A and the side surface 222B. Accordingly, since the contact area of the bonding portion 30 on the columnar portion 222 increases, the current density in the bonding layer 30 can be reduced. Accordingly, it is possible to further suppress the occurrence of electromigration in the bonding layer 30.

(52) In the semiconductor device A10, the columnar portions 222 of the electrodes 22 of the semiconductor element 20 are each provided with a recessed portion 222C that is recessed from the leading end surface 222A to the element main body 21. The recessed portions 222C are in contact with the bonding layer 30. Accordingly, an anchoring effect on the columnar portion 222 occurs in the bonding layer 30. For this reason, it is possible to achieve an improvement in the bond strength between the columnar portions 222 and the bonding layer 30.

(53) The semiconductor layer 212 of the element main body 21 of the semiconductor element 20 includes the switching circuit 212A. The multiple first electrodes 22A included in the multiple electrodes 22 are electrically connected to the switching circuit 212A. On the other hand, the rear surfaces 102 of the multiple first leads 11 that are included in the conductive members 10 and to which the multiple first electrodes 22A are electrically bonded are exposed from the bottom surface 42 of the sealing resin 40. Accordingly, when the semiconductor device A10 is used, the heat generated from the semiconductor element 20 due to the driving of the switching circuit 212A can be efficiently dissipated to the outside.

(54) As stated before, each of the multiple first electrodes 22A includes a base portion 221 and a columnar portion 222. The constituent material of the columnar portion 222 includes copper. The columnar portion 222 has a shorter length and a greater cross-sectional area than a bonding wire. For this reason, it is possible to reduce parasitic resistance between the first leads 11 and the switching circuit 212A compared to a case in which the first leads 11 and the base portion 221 are connected by bonding wires. If the parasitic resistance is reduced, an effect of reducing the on resistance and noise of the switching circuit 212A is obtained.

(55) Each of the multiple first leads 11 includes a main portion 111 that extends in the first direction x, and a pair of side portions 112 that are connected to the two ends in the first direction x of the main portion 111. The pair of side portions 112 each include a first end surface 112A that faces the first direction x and is exposed from the first side surface 431 of the sealing resin 40. The pair of first end surfaces 112A are flush with the first side surface 431. In the second direction y, the dimensions b of the pair of first side surfaces 112A are smaller than the dimensions B of the rear surfaces 102 of the main portions 111. Accordingly, the areas of the pair of first end portions 112A can be made smaller than the areas of the first end surfaces in a conventional QFN semiconductor device. For this reason, blade dicing is performed in the manufacture of the semiconductor device A10, the occurrence of metal burrs on the pair of first end surfaces 112A is

suppressed. If the occurrence of metal burrs is suppressed, it is possible to achieve an improvement in the mountability of the semiconductor apparatus **A10** on the wiring substrate.

(56) As shown in FIG. **9**, wedge portions **112B** is formed on each of the pair of side portions **112** of the multiple first leads **11** (the first input terminal **11A** and the output terminal **11C**). Accordingly, the dimensions **b** of the pair of first end surfaces **112A** can be made smaller in the second direction **y** than the dimensions **B** of the rear surfaces **102** of the main portions **111** of the first leads **11**. Also, the wedge portions **112B** are in contact with the sealing resin **40** in the first direction **x**.

Accordingly, it is possible to prevent the multiple first leads **11** from coming out from the pair of first side surfaces **431** of the sealing resin **40**.

(57) As shown in FIG. **10**, the notch portions **112C** are formed in the pairs of side portions **112** of the first leads **11** (second input terminal **11B**). Accordingly, the dimensions **b** of the pair of first end surfaces **112A** can be made smaller in the second direction **y** than the dimensions **B** of the rear surfaces **102** of the main portions **111** of the first leads **11**. The sealing resin **40** fills the notch portions **112C**. Accordingly, the first leads **11** are in contact with the sealing resin **40** in the first direction **x**. Accordingly, it is possible to prevent the multiple first leads **11** from coming out from the pair of first side surfaces **431** of the sealing resin **40**.

(58) The second input terminals **11B** include multiple protruding portions **113** that protrude from the other side in the second direction **y** of the main portions **111**. The multiple protruding portions **113** each have an auxiliary end surface **113A** that faces the second direction **y**. The multiple auxiliary end surfaces **113A** are exposed from the second side surface **432** of the sealing resin **40** located on the other side in the second direction **y**. Accordingly, the second input terminal **11B** is in contact with the sealing resin **40** on the other side in the second direction **y**. Thus, it is possible to prevent the second input terminal **11B** from coming out from the second side surface **432** located on the other side in the second direction **y**.

(59) In each of the multiple first leads **11**, the area of the main surface **101** is greater than the area of the rear surface **102**. Accordingly, the multiple first leads **11** are in contact with the sealing resin **40** on the side that the rear surface **102** faces in the thickness direction **z**. Accordingly, it is possible to prevent the multiple first leads **11** from coming out from the bottom surface **42** of the sealing resin **40**. Also, it is possible to increase the areas of the main surfaces **101** of the multiple first leads **11** to which the multiple first electrodes **22A** of the semiconductor element **20** are electrically bonded. This makes it possible to achieve an increase in the number of the multiple first electrodes **22A**.

(60) The conductive members **10** further include multiple second leads **12** to which at least some of the multiple second electrodes **22B** included in the multiple electrodes **22** are electrically bonded. In each of the multiple second leads **12**, the area of the main surface **101** is greater than the area of the rear surface **102**. Accordingly, similarly to the relationship between the main surfaces **101** and the rear surfaces **102** of the above-described first leads **11**, it is possible to prevent the multiple second leads **12** from coming out from the bottom surface **42** of the sealing resin **40**. Also, since it is possible to further increase the areas of the main surfaces **101** of the multiple second leads **12** to which the multiple second electrodes **22B** are electrically bonded, it is possible to achieve an increase in the number of the multiple second electrodes **22B**.

(61) A semiconductor device **A20** according to a second embodiment will be described with reference to FIGS. **17** and **18**. In these drawings, elements that are the same as or similar to those of the above-described semiconductor device **A10** will be denoted by the same reference numerals, and redundant description will be omitted. Here, the cross-section position of FIG. **17** is the same as the cross-section position of FIG. **15**. The cross-section position of FIG. **18** is the same as the cross-section position of FIG. **16**.

(62) In the semiconductor device **A20**, the configuration of the multiple electrodes **22** of the semiconductor element **20** is different from the configuration of the above-described semiconductor device **A10**.

(63) As shown in FIGS. 17 and 18, in each of the multiple electrodes 22 (the multiple first electrodes 22A and the multiple second electrodes 22B), the leading end surface 222A of the columnar portion 222 has a protruding shape that bulges toward the main surface 101 of the conductive member 10.

(64) Next, actions and effects of the semiconductor device A20 will be described.

(65) The semiconductor device A20 includes: conductive members 10 having main surfaces 101; a semiconductor element 20 having an element main body 21 and multiple electrodes 22 that are electrically bonded to the main surfaces 101; and a bonding layer 30 that is in contact with both the main surfaces 101 and the multiple electrodes 22. The multiple electrodes 22 each have a base portion 221 that is in contact with the side of the element main body 21 opposing the main surface 101, and a columnar portion 222 that protrudes from the base portion 221 to the main surfaces 101 and is in contact with the bonding layer 30. The bonding layer 30 is a sintered body in which metal powder is bonded together. Accordingly, with the semiconductor device A20 as well, it is possible to suppress the occurrence of electromigration in the bonding layer 30 used in flip-chip bonding.

(66) With the semiconductor apparatus A20, the leading end surfaces 222A of the columnar portions 222 have protruding shapes that bulge toward the main surfaces 101 of the conductive members 10 in the multiple electrodes 22. Accordingly, when the semiconductor element 20 is electrically bonded to the conductive members 10 through flip-chip bonding, the bonding layer 30 that is interposed between the main surfaces 101 and the columnar portions 222 is spread out in a direction orthogonal to the thickness direction z. The spread-out bonding layer 30 is in contact with the side surfaces 222B of the columnar portions 222. For this reason, the contact area of the bonding layer 30 on the side surface 222B further increases, and therefore it is possible to reduce the current density of the bonding layer 30. Accordingly, the occurrence of electromigration in the bonding layer 30 can be suppressed further than with the above-described semiconductor device A10.

(67) A semiconductor device A30 according to a third embodiment will be described with reference to FIGS. 19 and 20. In these drawings, elements that are the same as or similar to those of the above-described semiconductor device A10 will be denoted by the same reference numerals, and redundant description will be omitted. Here, the cross-section position of FIG. 19 is the same as the cross-section position of FIG. 15. The cross-section position of FIG. 20 is the same as the cross-section position of FIG. 16.

(68) In the semiconductor device A30, the configurations of the multiple electrodes 22 and the outer surface protection film 23 of the semiconductor element 20 are different from those in the above-described semiconductor apparatus A10.

(69) As shown in FIGS. 19 and 20, in each of the multiple electrodes 22 (the multiple first electrodes 22A and the multiple second electrodes 22B), the leading end surface 222A of the columnar portion 222 is parallel to the main surface 101 of the conductive member 10.

(70) As shown in FIGS. 19 and 20, the outer surface protection film 23 is spaced apart from the columnar portions 222 of the multiple electrodes 22. The outer surface protection film 23 is provided with multiple openings 231 that penetrate in the thickness direction z. The columnar portions 222 of the first electrodes 22A and the second electrodes 22B are exposed from the multiple openings 231. Accordingly, when forming the multiple electrodes 22, the volumes of the columnar portions 222 can be made larger than the volumes of the columnar portions 222 of the above-described semiconductor apparatus A10 and semiconductor apparatus A20.

(71) Next, actions and effects of the semiconductor device A30 will be described.

(72) The semiconductor device A30 includes: conductive members 10 having main surfaces 101; a semiconductor element having an element main body 21 and multiple electrodes 22 that are electrically bonded to the main surfaces 101; and a bonding layer 30 that is in contact with both the main surfaces 101 and the multiple electrodes 22. The multiple electrodes 22 each have a base portion 221 that is in contact with the side of the element main body 21 opposing the main surface

101, and a columnar portion **222** that protrudes from the base portion **221** to the main surfaces **101** and is in contact with the bonding layer **30**. The bonding layer **30** is a sintered body in which a metal powder is bonded together. Accordingly, with the semiconductor device **A30** as well, it is possible to suppress the occurrence of electromigration in the bonding layer **30** used in flip-chip bonding.

(73) In the semiconductor device **A30**, the outer surface protection film **23** of the semiconductor element **20** is spaced apart from the columnar portions **222** of the multiple electrodes **22**.

Accordingly, the volumes of the columnar portions **222** can be made greater than the volumes of the columnar portions **222** of the above-described semiconductor device **A10** and semiconductor device **A20**. For this reason, the contact area of the bonding layer **30** on the columnar portion **222** further increases, and therefore it is possible to reduce the current density of the bonding layer **30**. Accordingly, the occurrence of electromigration in the bonding layer **30** can be suppressed further than with the above-described semiconductor device **A10**.

(74) In the examples indicated by the semiconductor device **A10**, the semiconductor device **A20**, and the semiconductor device **A30**, the conductive members **10** are the multiple leads (the multiple first leads **11**, the multiple second leads **12**, and the pair of third leads **13**) that are formed by the same lead frame. In another configuration of the conductive member **10**, it is also possible to include an insulating substrate, and a conductive layer including multiple regions that are arranged on the insulating substrate and are spaced apart from each other.

(75) The present disclosure is not limited to the above-described embodiments. Specific configurations of the units of the present disclosure can be designed and modified in various ways.

Claims

1. A semiconductor device comprising: an electric conductor having a main surface and a rear surface that face mutually opposite sides in a thickness direction; a semiconductor element including an element main body and a plurality of electrodes, the element main body having a first side facing the main surface of the electric conductor, the plurality of electrodes each protruding toward the main surface from the first side of the element main body and being electrically connected to the main surface; and a bonding layer held in contact with the main surface and the plurality of electrodes, wherein each of the plurality of electrodes includes: a base portion in contact with the element main body; and a columnar portion protruding toward the main surface from the base portion and held in contact with the bonding layer, and wherein the electric conductor comprises: a plurality of first leads that extend in a first direction orthogonal to the thickness direction and are separated from each other in a second direction orthogonal to the thickness direction and the first direction; and a plurality of second leads that are located offset from the plurality of first leads in the second direction, the semiconductor element includes a semiconductor substrate and a semiconductor layer disposed on the semiconductor substrate and facing the main surface, the semiconductor layer is formed with a switching circuit and a control circuit electrically connected to the switching circuit, the plurality of electrodes include a plurality of first electrodes and a plurality of second electrodes, the plurality of first electrodes are electrically connected to the switching circuit and are electrically connected to the plurality of first leads, and the plurality of second electrodes are electrically connected to the control circuit, and at least part of the plurality of second electrodes is electrically connected to the plurality of second leads.

2. The semiconductor device according to claim 1, wherein the columnar portion of each electrode includes a leading end surface facing the main surface and a side surface connected to the leading end surface and facing in a direction orthogonal to the thickness direction, and the bonding layer is in contact with the leading end surface and the side surface.

3. The semiconductor device according to claim 2, wherein the semiconductor element includes an

outer surface protection film that covers the first side of the element main body, and the leading end surface of each electrode is located between the main surface and the outer surface protection film in the thickness direction.

4. The semiconductor device according to claim 3, wherein the outer surface protection film is in contact with the base portion and the columnar portion of each electrode.

5. The semiconductor device according to claim 3, wherein the outer surface protection film is spaced apart from the columnar portion of each electrode.

6. The semiconductor device according to claim 4, wherein the columnar portion of each electrode is provided with a recessed portion that is recessed toward the element main body from the leading end surface, and the bonding layer is in contact with the recessed portion.

7. The semiconductor device according to claim 4, wherein the leading end surface of each electrode has a protruding shape that bulges toward the main surface.

8. The semiconductor device according to claim 4, wherein the base portion of each electrode is made of a material including aluminum.

9. The semiconductor device according to claim 4, wherein the base portion of each electrode is made of a material including copper.

10. The semiconductor device according to claim 1, further comprising a sealing resin that covers a part of each first lead, a part of each second lead and the semiconductor element, wherein the sealing resin has: a bottom surface that faces a same side as the rear surface in the thickness direction; and a pair of first side surfaces that are connected to the bottom surface and are spaced apart from each other in the first direction, the plurality of first leads each include a main portion that extends in the first direction and a pair of side portions that are connected to ends of the main portion in the first direction, each of the pair of side portions has a first end surface that faces in the first direction, the rear surface of each first lead is exposed from the bottom surface, the first end surfaces are exposed from the pair of first side surfaces so as to be flush with the first side surfaces, and a dimension in the second direction of the pairs of first end surfaces is smaller than a dimension in the second direction of the rear surfaces of the main portions.

11. The semiconductor device according to claim 10, wherein each of the pair of side portions is provided with wedge portions that extend from the main surface to the rear surface and are recessed inward of the side portions from both sides in the second direction.

12. The semiconductor device according to claim 10, wherein each of the pair of side portions is provided with a notch portion that extends from the main surface to the rear surface, is recessed in the first direction from the first end surface, and divides the first end surface into two regions.

13. The semiconductor device according to claim 10, wherein each of the plurality of second leads includes a second end surface facing in the second direction, the sealing resin includes a pair of second side surfaces that are connected to the bottom surface and the pair of first side surfaces and are spaced apart from each other in the second direction, the rear surfaces of the plurality of second leads are exposed from the bottom surface, and the plurality of second end surfaces are exposed from one of the second side surfaces so as to be flush with the one of the second side surfaces.

14. The semiconductor device according to claim 13, wherein one of the plurality of first leads is formed with a plurality of protruding portions that protrude from the main portion of the one of the plurality of first leads, each of the plurality of protruding portions includes an auxiliary end surface that faces in the second direction, and the auxiliary end surface of each protruding portion is exposed from the other of the second side surfaces so as to be flush with the other of the second side surfaces.
